Below is a comprehensive, professional Model Risk Management (MRM) documentation template you can use for any model—AI/ML, statistical, financial, or operational. It aligns with common regulatory expectations (e.g., OCC 2011-12 / SR 11-7–style frameworks) and modern AI governance standards.

Model Risk Management Documentation

Summary: This document describes the Model Risk Management (MRM) framework for the Proactive Measles Detector, a supervised machine learning model designed to identify U.S. counties at risk of measles outbreaks. The model leverages demographic, vaccination, and behavioral data to generate county-level risk scores and classifications. Given its potential influence on public health decision-making and resource allocation, the model is classified as high-risk and is subject to rigorous validation, monitoring, and governance standards. The framework prioritizes recall to minimize missed outbreaks and supports explainable, actionable insights for public health stakeholders.

1. Model Overview

1.1 Model Name: Proactive Measles Detector

1.2 Model ID / Version: M_Detector_v1

1.3 Date Created / Last Updated:

Created on March 30th, 2026

Last updated on April 20th, 2026

1.4 Model Owner: Owen Williamson

1.5 Model Developer(s): Owen Williamson

1.6 Model Reviewer(s): Owen Williamson

1.7 Model Users / Stakeholders:

County officials

State health departments

1.8 Purpose & Business Use Case

This model identifies counties currently experiencing measles outbreaks and, more importantly, those at elevated risk of future outbreaks based on demographic, vaccination, and behavioral characteristics.

The primary objective is proactive intervention. By identifying high-risk counties before outbreaks occur, public health officials can deploy targeted strategies such as vaccination campaigns (MMR), localized awareness initiatives, and resource allocation.

Fortunately, there have been no reported deaths due to measles outbreaks in the United States. However, measles outbreaks can impose significant financial burdens on tax payers and health departments. According to Johns Hopkins estimates, the initial case in an outbreak can cost approximately $250,000, with additional cases costing around $15,000 each. Early detection and intervention therefore provide both public health and economic benefits.

This model supports decision-making for:

- County-level intervention prioritization
- Resource allocation planning
- Preventative healthcare strategy development

1.9 Model Category / Type

- Primary Models:
    - Supervised classification models: Logistic Regression, Balanced Random Forest, and XGBoost.
    - Logistic Regression is used as a transparent baseline model
    - Balanced Random Forest is used to capture nonlinear relationships while addressing class imbalance.
    - XGBoost is the primary predictive model due to its strong performance and ability to model complex interactions.

The final modeling framework focuses only on supervised classification for county-level measles outbreak risk detection and transmission risk assessment.

1.10 Criticality / Risk Tiering

- High: Rationale
    - Direct public health impact (potential loss of life)
    - Influences allocation of limited healthcare and public health resources
    - Risk of false negatives (missed potential outbreaks) has higher consequences

        - Model outputs may influence government decision making
- Therefore, high standards for validation, monitoring and governance are required.

---

2. Model Scope & Boundaries

2.1 In-Scope Components

Describe components included in the model boundary:

- Data pipeline
    - Integration of CDC, US Census and John Hopkins datasets
- Feature engineering
    - Missing value imputation at the county/state level
    - One-hot encoding for categorical variables
    - Correlation-based feature reduction
    - Model-based feature selection
    - Features created that mix two variables together to get interaction effects and greater interpretation from the models.
- Algorithm(s)
    - Logistic regression (baseline, interpretable benchmark)
    - Random forest (nonlinear relationship with imbalance handling)
    - XGBoost (primary predictive model)
- Post-processing
    - Probability threshold tuning for classification
        - Comparison of alternative outbreak target definitions:
            - Target_At_Least_1: counties with one or more cases
            - Target_At_Least_2: counties with two or more cases
                - This indicates local transmission
    - SHAP-based global and local explainability
    - Identification of high-risk counties without current reported outbreaks
- Monitoring dashboards
    - Streamlit dashboard for visualization and stakeholder use

2.2 Out-of-Scope Components

Document excluded systems or processes.

- Individual-level prediction of measles cases
- Real-time outbreak tracking beyond available data refresh frequency
- Vaccine distribution logistics optimization
- Clinical diagnosis or medical decision-making

- International measles outbreak prediction outside the United States
- Integration with electronic health record (EHR) systems

2.3 Dependencies & Interfaces

- Upstream data sources
    - Vaccine Hesitancy - US Census
    - Measles Cases by County - John Hopkins
    - CDC MMR Vaccine Coverage - CDC
    - County Demographics - US Census
- Downstream consumers
    - State and county health officials via dashboard
- APIs, databases, platforms
    - Positron for Python coding
    - Streamlit for dashboard

---

3. Data Documentation

3.1 Data Sources

List each source with:

- Description
- Owner
- Refresh frequency
- Access controls

John Hopkins Measles Cases by County Tracker

- Owner: John Hopkins, Refreshed ~ Once Weekly, Access Available via Github Repository

Vaccine Hesitancy for COVID 19

- Owner: CDC, Not Regularly Refreshed, Available via CDC Gov page

US Census County Characteristics

- Owner: US Census, Updated Every 5 Years, Available via US Census gov page

CDC MMR Vaccine Coverage

- Owner: CDC, Estimated Every School Year, Available via data.cdc.gov

3.2 Data Quality Assessment

- Completeness
    - County-level coverage is strong across the merged dataset, but some variables contain missing values, especially vaccination- and hesitancy-related fields. Missingness was most common in state-reported vaccination coverage and selected demographic or vulnerability variables. To preserve geographic structure, missing numeric fields were imputed using state-level medians, with fallback to the global median when needed.
- Accuracy
    - Data sources were obtained from reputable public institutions, including the U.S. Census, CDC, and Johns Hopkins measles tracker. However, accuracy is still limited by the quality of the source systems. Measles case counts may be affected by underreporting, delayed reporting, or inconsistent county attribution. Vaccine hesitancy is estimated rather than directly observed, and COVID-19 vaccine hesitancy is used as a proxy for broader vaccine resistance behavior.
- Timeliness
    - John Hopkins provides close to real-time data and provides a comprehensive count of reported measles cases
    - Vaccine hesitancy is from 2021 and is used as proxy; it would be more favorable if we had up-to-date metrics.
    - Overall, the dataset is appropriate for county-level risk modeling, but not for real-time operational outbreak surveillance.
- Consistency
    - The major consistency issue involved differences in county identifiers and formatting across sources. FIPS codes had to be standardized to a common 5-digit format before merging. State names and geography labels also required alignment across vaccination and census files. Percentage variables appeared in mixed formats and were standardized before modeling.
- Known data issues / mitigations
    - FIPS codes required zero-padding and type conversion before merging.
    - County and state naming conventions varied across datasets and required standardization.
    - Percentage fields were stored as strings in some files and converted to numeric decimal form.
    - Missing numeric values were handled using state-level median imputation.
    - Target leakage risk from raw case-count fields was mitigated by removing total_measles_cases and related direct identifiers from modeling features.

3.3 Data Preprocessing

- Cleaning steps
    - Standardized FIPS codes to 5-digit strings across all source datasets.
    - Removed rows corresponding to state-level aggregates where county-level modeling was required.
    - Filtered vaccination data to MMR-specific records only.
    - Selected the most recent available vaccination record by geography.
    - Cleaned percentage fields by removing percent signs and converting them to numeric values.
    - Standardized rate features onto a 0–1 scale when appropriate.
- Transformations
    - Converted vaccination and hesitancy percentages into numeric rates.
    - Created grouped demographic variables such as AGE_18_44, AGE_45_PLUS, and AGE_CHILD.
    - Created ratio features such as AGE_18_44_RATIO, AGE_45_PLUS_RATIO, and AGE_CHILD_RATIO.
    - Created engineered interaction terms, including:
        - unvaccinated_rate
        - low_vax_flag
        - hesitancy_vax_interaction
        - risk_interaction
        - risk_interaction_2
        - vax_hesitancy_interaction
        - not_covid_vaccinated
- Filtering rules
    - Removed direct target-like variables such as total measles case counts from the model input.
    - Removed identifiers and administrative fields such as FIPS, county names, and state codes from the final modeling matrix.
    - Removed geometry and map-shape fields.
    - Removed race, ethnicity, and overly granular demographic proxy fields where they were not necessary for the final model.
    - Removed highly redundant or overlapping raw age fields after grouped age variables were constructed.
- Handling of missing values
    - Missing numeric values were imputed using state-level medians to preserve geographic structure. If a state-level median could not be computed, the global median was used. After feature engineering, a second pass ensured that remaining numeric missingness and divide-by-zero artifacts were addressed.
- Encoding / normalization
    - Low-cardinality categorical variables were one-hot encoded.
    - High-cardinality categorical variables were dropped rather than expanded.
    - Logistic Regression used scaled features.
    - Tree-based models used unscaled features, though the preprocessing pipeline maintained consistent numeric formatting across all models.

3.4 Training / Testing / Validation Samples

- Time ranges: The merged dataset combines the most recent available county demographic estimates, historical vaccination coverage, COVID-19 hesitancy estimates, and current measles case tracking. Because source datasets come from different years, the modeling task is cross-sectional rather than a true time-series forecast.
- Sample sizes: The final county-level dataset includes roughly all U.S. counties with available mergeable information. The dataset was split into:
    - 70% training
    - 15% validation
    - 15% test

  Exact sample counts vary slightly depending on the target definition and post-filtering preprocessing steps.

- Methods for splitting: A stratified random split was used so that positive outbreak counties remained proportionally represented across training, validation, and test samples. This was especially important because both target definitions are imbalanced, and the Target_At_Least_2 formulation is even more sparse than Target_At_Least_1.
- Representativeness analysis: The dataset is geographically broad and covers counties across the United States. However, positive cases are rare, which means the training set contains relatively few counties with outbreaks. The Target_At_Least_1 definition is better for broad detection, while Target_At_Least_2 is more representative of local transmission but creates a harder, more imbalanced learning problem. Class imbalance mitigation was therefore necessary in model design.

---

4. Methodology & Model Design

4.1 Conceptual Design

A supervised classification framework was used to balance predictive accuracy and interpretability.

Logistic Regression, Balanced Random Forest, and XGBoost were used to estimate the probability of measles outbreak risk at the county level. Two target definitions were evaluated:

- Target_At_Least_1: counties with one or more reported measles cases
- Target_At_Least_2: counties with two or more reported measles cases

This two-target framework allows the model to distinguish between general exposure risk (Target_At_Least_1) and more meaningful local transmission risk (Target_At_Least_2). The first target is useful for early-warning surveillance, while the second target better reflects sustained outbreak conditions within a county.

The final framework supports proactive intervention by identifying counties with demographic, behavioral, and vaccination-related characteristics associated with elevated measles risk.

4.2 Algorithm Selection Rationale

Multiple supervised models were selected to balance interpretability, predictive performance, and robustness:

- Logistic Regression: provides a transparent benchmark and interpretable baseline
- Balanced Random Forest: captures nonlinear relationships and interactions while handling class imbalance
- XGBoost: selected as the primary model due to strong performance on imbalanced classification tasks, threshold flexibility, and superior explainability when paired with SHAP

KNN and clustering-based approaches were not retained in the final project because they provided less actionable value than the supervised framework and reduced the conceptual clarity of the modeling objective.

4.3 Model Architecture

Include diagrams if applicable.

```
[Data Sources]
 (CDC | Census | JHU)
        ↓
[Data Cleaning & Merge]
        ↓
[Feature Engineering]
 (ratios, interactions)
        ↓
[Models]
 (LogReg | RF | XGB)
        ↓
[Threshold Tuning]
        ↓
[Predictions]
 (Risk Score + Class)
        ↓
[Dashboard]
 (Map + SHAP)
```

End-to-End Modeling Pipeline for Measles Risk Detection

The final modeling architecture is a supervised county-level classification pipeline with shared preprocessing and multiple model classes.

Pipeline stages:

1. Data ingestion from census, measles, vaccination, and hesitancy sources
2. Identifier standardization and merging
3. Numeric cleaning and percentage normalization
4. State-level median imputation
5. Feature engineering and grouped demographic construction
6. Leakage and identifier removal
7. One-hot encoding of remaining low-cardinality categorical variables
8. Model-based feature selection using Random Forest importance
9. Training of benchmark and nonlinear classifiers
10. Validation-set threshold tuning for XGBoost
11. Test-set evaluation
12. SHAP-based explainability on the held-out test set

Final model stack:

- Logistic Regression: benchmark and interpretable baseline
- Balanced Random Forest: nonlinear benchmark with imbalance handline
- XGBoost: final champion model

Two binary target definitions were evaluated:

- Target_At_Least_1: counties with one or more measles cases
- Target_At_Least_2: counties with two or more measles cases

This allows the architecture to support both early detection and sustained-transmission risk assessment.

4.4 Feature Engineering

- List features
    - unvaccinated_rate
    - low_vax_flag
    - not_covid_vaccinated
    - hesitancy_vax_interaction
    - risk_interaction
    - risk_interaction_2
    - vax_hesitancy_interaction
    - AGE_18_44_RATIO
    - AGE_45_PLUS_RATIO

    - AGE_CHILD_RATIO
- Definitions
    - **unvaccinated_rate**: 1 - Estimate (%); represents the county-level gap in MMR coverage.
    - **low_vax_flag**: binary indicator equal to 1 when MMR vaccination is below 90%; used as a herd-immunity warning signal.
    - **not_covid_vaccinated**: 1 - Percent adults fully vaccinated against COVID-19; used as a proxy for broader vaccine resistance or low adult vaccine uptake.
    - **hesitancy_vax_interaction**: interaction between estimated hesitancy and MMR unvaccinated rate.
    - **risk_interaction**: interaction between child population share and hesitancy, capturing heightened risk when susceptible age groups coexist with resistance to vaccination.
    - **risk_interaction_2**: interaction between social vulnerability and hesitancy, capturing access and trust barriers.
    - **vax_hesitancy_interaction**: interaction between adult COVID vaccination and hesitancy, intended to reflect generalized vaccine behavior patterns.
    - **AGE_*_RATIO features**: grouped age totals normalized by county population to better compare counties of different sizes and reduces redundancy in age brackets.
- Importance
    - These help to add greater context for the model and for interpretability, and add needed information for the models.
    - Feature engineering was used to improve both predictive performance and interpretability.
    - The grouped age variables made the model easier to explain and reduced redundancy across raw demographic bins.
    - Interaction terms were especially useful in tree-based models because they exposed nonlinear relationships between hesitancy, age structure, and vulnerability.
- Transformations
    - Percent-based features were converted to numeric rates.
    - Highly overlapping raw age columns were collapsed into grouped age measures.
    - Ratio features were calculated using county population as the denominator.
    - Infinite values created by division were replaced and re-imputed.
- Rationale for inclusion/exclusion: Included features were selected because they reflect plausible outbreak drivers: vaccination coverage, vaccine resistance, age structure, and social vulnerability. Excluded features generally fell into one of the following groups:
    - direct leakage fields
    - identifiers
    - geometry/map fields
    - very high-cardinality categoricals
    - redundant raw demographic detail that was replaced by grouped measures

4.5 Model Assumptions

- Statistical assumptions
    - Counties are treated as independent observations.
    - The supervised classification problem is cross-sectional, not longitudinal.

    - Nonlinear relationships are expected and are handled by tree-based models.
    - Class imbalance is substantial and must be explicitly addressed in modeling and evaluation.
    - Validation data can be used to select classification thresholds, which are then fixed before test evaluation.
- Business assumptions
    - For public health, we would rather over-predict potential counties for an outbreak than to miss a county that has an outbreak. Overall, we want Recall to be higher so that we get positive cases, and allow for false positives in classification models.
    - Missing a true outbreak is more costly than flagging a false alarm.
    - Public health users can act on ranked or thresholded county risk predictions.
    - County-level risk is a meaningful planning unit for intervention prioritization.
    - Two target definitions represent two practical use cases:
        - Target_At_Least_1 for early detection
        - Target_At_Least_2 for sustained transmission risk
- Data assumptions
    - COVID-19 vaccination hesitancy is an acceptable proxy for broader vaccine resistance behavior.
    - County demographic structure is stable enough to be useful in a cross-sectional model.
    - The merged data sources, although temporally imperfect, still capture meaningful county-level risk structure.
    - Reported measles cases are directionally informative even if some underreporting exists.

4.6 Limitations

- Technical:
    - Class imbalance remains a significant challenge, especially for the stricter Target_At_Least_2 definition
    - Performance depends on the selected threshold, particularly when prioritizing recall versus precision
    - Some engineered features are correlated and may split importance across related variables
- Data:
    - Potential underreporting of measles cases
    - Temporal misalignment between datasets
    - Missing or noisy vaccination data
    - COVID-19 vaccine hesitancy used as proxy for MMR vaccine hesitancy
- Business:
    - Model predictions are probabilistic and should not be used as sole decision criteria
- Ethical:
    - Risk of misallocation of resources if predictions are incorrect
    - Potential bias due to demographic proxies in data

---

5. Model Performance

5.1 Training Performance Metrics

e.g., accuracy, AUC, RMSE, precision/recall, stability.

Primary performance metric was Recall and ROC-AUC for good detection of outbreaks, while allowing for False Positives to indicate counties with higher risk of potential outbreak of measles.

Primary evaluation metrics included Recall, Precision, F1 Score, and ROC-AUC.

Recall was prioritized due to the high cost of false negatives (missed outbreaks).
Precision was monitored to ensure operational feasibility and avoid excessive false positives.

F1 Score was used to balance precision-recall tradeoffs.
ROC-AUC was used to evaluate overall model discrimination ability across thresholds.

5.2 Validation Performance

Model performance was evaluated on a holdout validation dataset for both target definitions.

- Logistic Regression provided an interpretable benchmark
- Balanced Random Forest provided strong nonlinear performance
- XGBoost consistently performed best overall and was selected as the primary model

Two target definitions were compared:

- Target_At_Least_1 produced better recall and is more suitable for early-warning detection
- Target_At_Least_2 produced higher ROC-AUC and cleaner interpretability, but lower recall, making it more suitable for identifying sustained transmission risk

Special emphasis was placed on recall to minimize false negatives, while threshold tuning was used to manage the tradeoff between sensitivity and precision.

5.3 Backtesting (if relevant)

- Method: A traditional backtest was not possible in the strict time-series sense because the final project dataset is assembled from cross-sectional public data sources collected at different times. Instead, out-of-sample validation and held-out test evaluation were used to simulate forward-looking performance.
- Results
    - Both target definitions were evaluated on held-out data:

        - Target_At_Least_1 produced stronger recall and broader detection
        - Target_At_Least_2 produced better rank-order separation and cleaner interpretability but lower recall
- Thresholds: XGBoost classification thresholds were optimized on the validation set using precision-recall tradeoffs and then fixed for test-set evaluation. This allowed operating points to better reflect public health priorities than the default 0.5 threshold.

5.4 Sensitivity Analysis

- Stress tests
    - The model was stress-tested conceptually by comparing two target definitions:
        - one or more cases
        - two or more cases
    - This revealed meaningful changes in feature importance and operational behavior.
- Adversarial tests: adversarial testing was not conducted. However, robustness was assessed by:
    - comparing multiple model classes
    - checking whether key explanatory variables remained stable across target definitions
    - reviewing SHAP results for consistency
- Scenario analysis: Scenario analysis showed that the Target_At_Least_1 formulation behaves like an early-warning screen, while the Target_At_Least_2 formulation behaves like a sustained-transmission detector. This supports the use of the two targets as complementary modeling perspectives rather than as interchangeable problem statements.

5.5 Benchmarking

Alternative models:

- Logistic Regression
- Balanced Random Forest
- XGBoost

Champion/challenger comparisons:

- Logistic Regression served as the transparent benchmark
- Balanced Random Forest served as the nonlinear challenger
- XGBoost was selected as the champion model

The project also benchmarked two target definitions:

- Target_At_Least_1 for broad outbreak detection
- Target_At_Least_2 for sustained, local transmission detection

The final project therefore benefits from presenting both as complementary views of outbreak risk.

5.6 Actionable Insights

The supervised modeling framework identified vaccination behavior and demographic structure as the primary drivers of predicted measles risk.

Key actionable findings include:

- Counties with higher proportions of unvaccinated individuals are consistently higher risk
- Counties with larger working-age populations (18–44) and higher child population ratios show elevated transmission risk; this indicates demographics play a significant role in transmission, especially looking at results for modeling with Target_At_Least_2, which would suggest local transmission within the county.
- Vaccine hesitancy amplifies underlying risk, particularly through interaction effects with vaccination and social vulnerability measures

These outputs support targeted public health strategies such as localized MMR vaccination outreach, education campaigns, and resource prioritization.

---

6. Explainability & Interpretability

6.1 Global Explainability

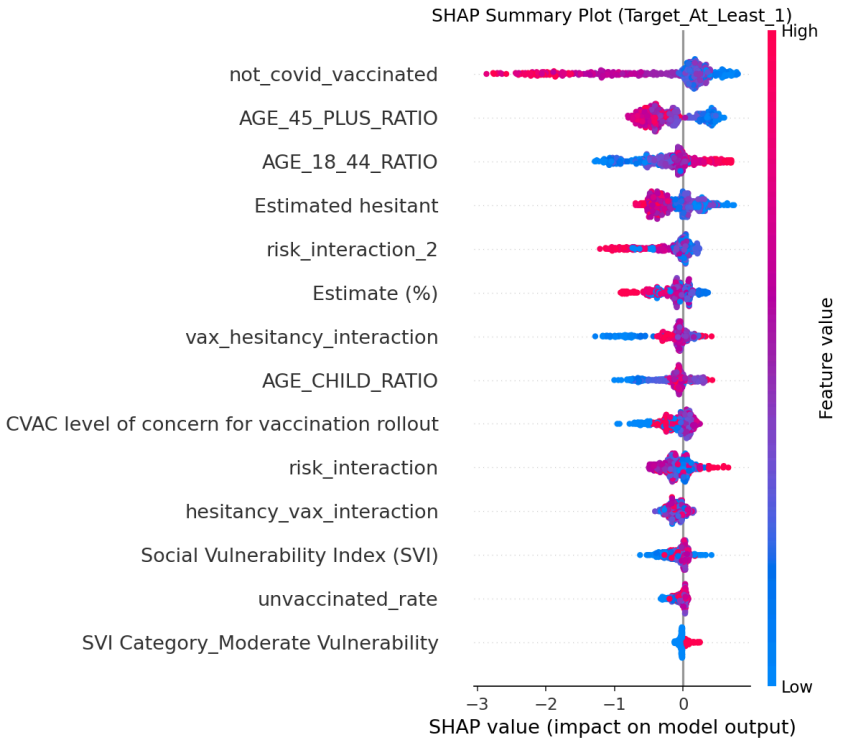
Global importance was assessed using:

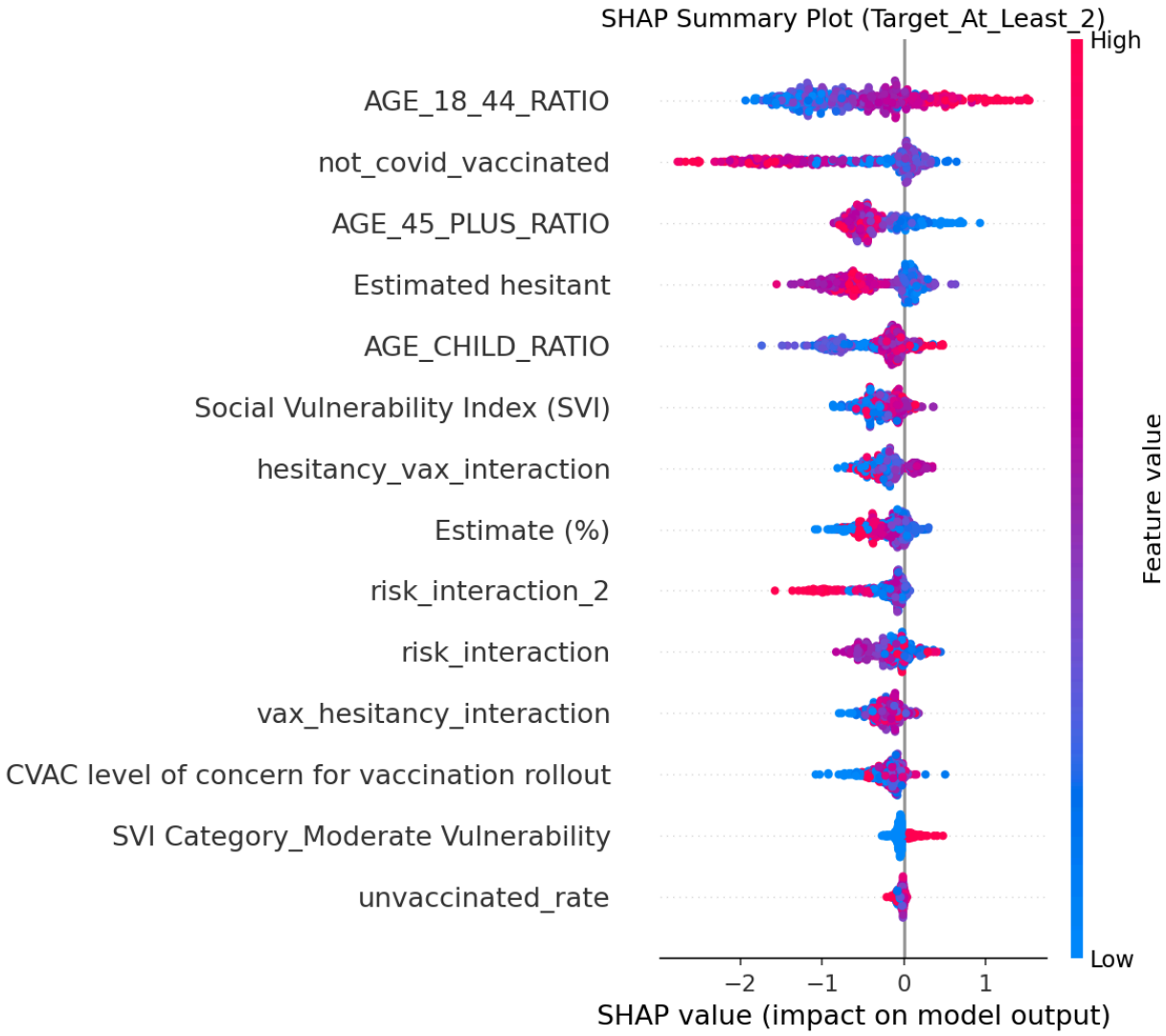
- SHAP summary plots
- SHAP bar plots

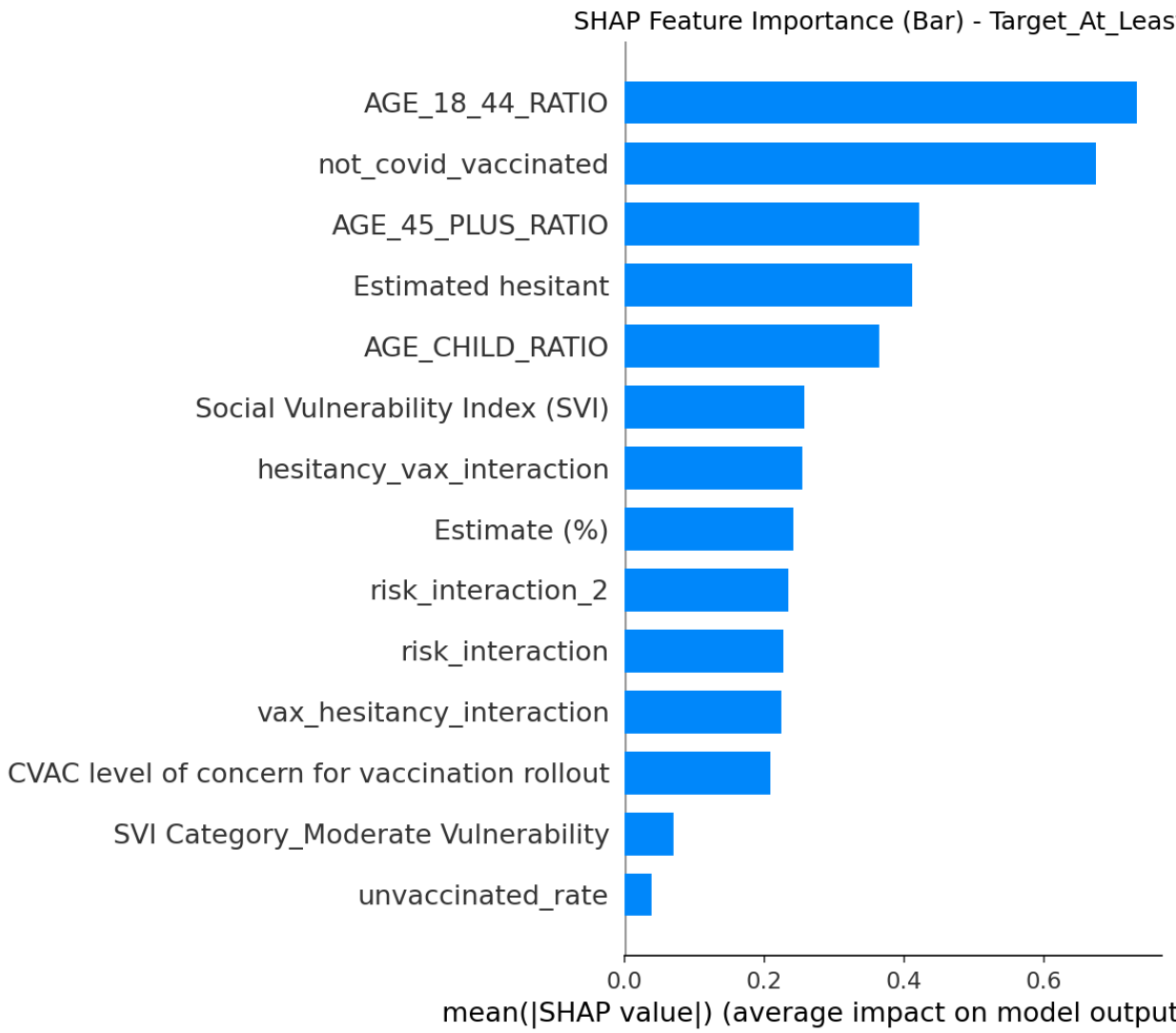
Across both target definitions, the dominant features were consistently tied to:

- age structure
- vaccination uptake
- adult non-vaccination proxy behavior
- vaccine hesitancy
- social vulnerability interactions

For Target_At_Least_1, vaccination-related features were especially dominant. For Target_At_Least_2, age-structure variables such as AGE_18_44_RATIO became even more prominent, suggesting that transmission risk is more dependent on population structure than isolated-case occurrence.


SHAP Summary Plot (Target_At_Least_1)
not_covid_vaccinated
AGE_45_PLUS_RATIO
AGE_18_44_RATIO
Estimated hesitant
risk_interaction_2
Estimate (%)
vax_hesitancy_interaction
AGE_CHILD_RATIO
CVAC level of concern for vaccination rollout
risk_interaction
hesitancy_vax_interaction
Social Vulnerability Index (SVI)
unvaccinated_rate
SVI Category_Moderate Vulnerability
−3
−2
−1
0
SHAP value (impact on model output)
High
Feature value
Low

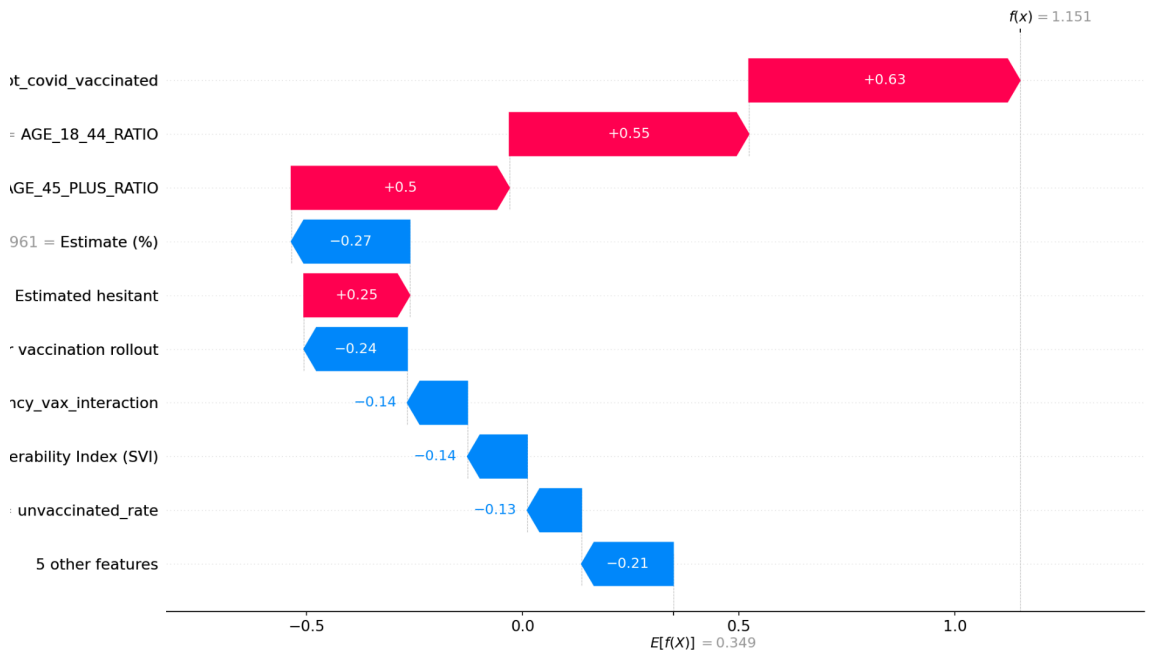

SHAP Summary Plot (Target_At_Least_2)
AGE_18_44_RATIO
not_covid_vaccinated
AGE_45_PLUS_RATIO
Estimated hesitant
AGE_CHILD_RATIO
Social Vulnerability Index (SVI)
hesitancy_vax_interaction
Estimate (%)
risk_interaction_2
risk_interaction
vax_hesitancy_interaction
CVAC level of concern for vaccination rollout
SVI Category_Moderate Vulnerability
unvaccinated_rate
−2
−1
0
1
SHAP value (impact on model output)
High
Low
Feature value



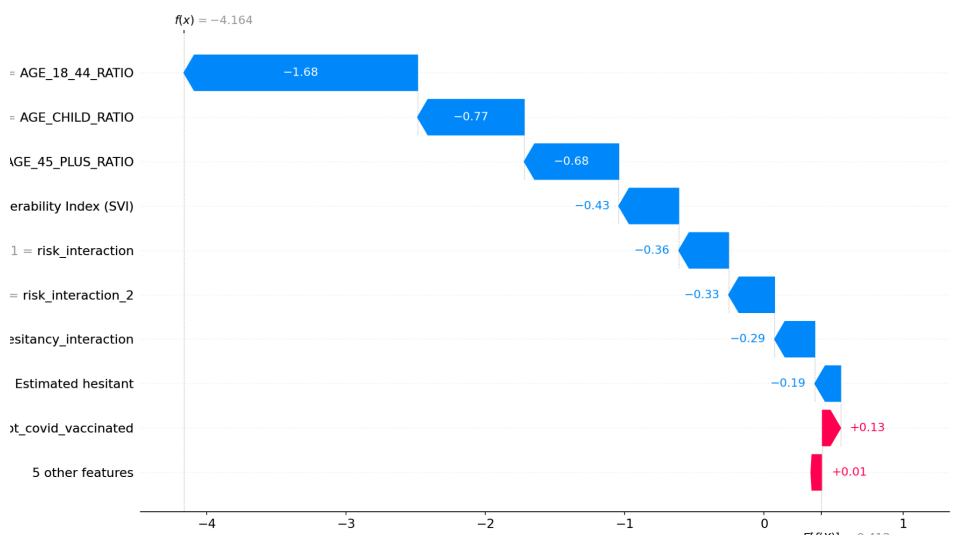
- Model reasoning
  - The model learns that counties are higher risk when they combine:
    - lower vaccination coverage
    - greater vaccine hesitancy
    - larger working-age and child population shares
    - higher social vulnerability in interaction with hesitancy
  - This pattern is epidemiologically reasonable because measles spread depends on both susceptibility and social contact structure.
- Explainability tools (e.g., SHAP, LIME)
  - SHAP was used as the primary explainability framework because it provides:
    - global ranking of influential features
    - directional interpretation of feature effects
    - local explanations for individual county predictions

6.2 Local Explainability

- Example predictions explained



  - This is for test sample using XGboost for general outbreak detection (Target_At_Least_1)
  - This shows a case that has been classified as 1, high probability of outbreak, where not_covid_vaccinated and age_18_44_ratio is driving the probability of measles outbreak higher.



  - This is for test sample using XGBoost for disease transmission (Target_At_Least_2)
  - This shows a case that has been classified as 0, not at higher risk for local transmission, where age demographics are driving the probability down

- Edge-case behavior

- The two target definitions revealed useful edge-case differences:
  - Under Target_At_Least_1, counties could be flagged based on vaccination-risk structure even if they did not resemble sustained transmission environments.
  - Under Target_At_Least_2, the model required stronger demographic and interaction evidence before classifying a county as positive.
- This suggests that local explanations are more conservative and transmission-focused under the stricter target definition.

6.3 Business Interpretability Assessment

Is the model understandable to stakeholders?

Model outputs are interpretable at a high level through:

- Risk scores (predicted probabilities)
- Binary risk classifications based on optimized thresholds
- SHAP-based explanations for both global and county-level predictions

These outputs can be communicated to non-technical stakeholders as:

- Higher-risk counties
- Lower-risk counties
- Counties at risk of sustained transmission

This enables actionable decision-making without requiring deep technical understanding.

---

7. Fairness, Bias, and Ethical Assessment

7.1 Protected Attributes Reviewed

The model incorporates county-level demographic and socioeconomic data, which may indirectly reflect protected characteristics. While individual-level protected attributes are not explicitly included, the following proxies were reviewed for potential bias:

- Race and ethnicity composition (from U.S. Census data)
- Income levels and poverty rates (SVI)
- Geographic location (urban vs. rural)

These variables may act as proxies for protected classes and were carefully evaluated to ensure the model does not disproportionately disadvantage specific populations.

7.2 Bias Metrics Evaluated

To assess fairness, model performance was evaluated across different population segments using the following metrics:

- Equal Opportunity: Comparison of true positive rates (recall) across groups to ensure equal detection of outbreaks
- Predictive Parity: Comparison of precision across groups to evaluate consistency in prediction accuracy

Subgroup review focused on potential proxy dimensions such as:

- social vulnerability levels
- urban versus rural structure where inferable
- demographic composition proxies available at the county level

Because the project uses county-level rather than individual-level data, fairness interpretation is necessarily approximate. The review therefore focused less on formal compliance testing and more on whether the model's error distribution and reliance on proxy variables suggested meaningful risk of systematic disadvantage.

No evidence of severe disparity was observed across reviewed proxy groups; however, continued monitoring is recommended due to the use of aggregated demographic features.

7.3 Fairness Mitigation Approaches

- Several strategies were implemented to mitigate potential bias:
    - Pre-processing:
        - Careful feature selection to remove redundant or highly correlated variables that may amplify bias
        - Handling of missing data to avoid systematic exclusion of certain populations
    - In-model:
        - Use of ensemble methods (Random Forest, XGBoost) to reduce reliance on any single biased feature
        - Class imbalance handling (e.g., class weights, scale_pos_weight) to ensure minority outbreak cases are properly detected
    - Post-processing:
        - Threshold tuning to prioritize recall, ensuring outbreaks are not disproportionately missed in vulnerable populations
        - SHAP-based review of feature effects to confirm that predictions are driven by meaningful risk factors rather than administrative artifacts

7.4 Ethical Considerations

List potential harms and mitigations.

Potential Harms:

- Misclassification Risk: False negatives may result in missed outbreaks, leading to delayed interventions and increased public health risk
- Resource Misallocation: False positives may lead to inefficient allocation of limited healthcare resources
- Bias Amplification: Use of demographic proxies may unintentionally reinforce existing inequalities in healthcare access

Mitigation Strategies:

- Prioritization of recall to minimize missed outbreaks
- Use of multiple supervised model classes and alternative target definitions to validate robustness of risk identification
- Transparent communication of model limitations to stakeholders
- Use of the model as a decision-support tool rather than a sole decision-maker

Ethical Use:

- The model is intended to support equitable public health interventions by identifying at-risk populations early, with the goal of improving access to preventative care such as vaccinations and education campaigns.

---

8. Model Controls & Governance

8.1 Access & Permission Controls

Who can retrain, update, deploy, or view outputs.

- Retrain: Model developer (Owen Williamson)

- Update: Model developer via controlled updates to preprocessing, feature engineering, and modeling code
- Deploy: Model developer through the Streamlit application
- View outputs: County officials, state health departments, and approved project stakeholders via dashboard outputs and summary reports

8.2 Change Management

- Versioning: First Version
- Code repositories
    - Github Repository: owilli38/6156-Final-Project-Measles-Detection-Model
- Approval requirements: N/a

Versioning follows a controlled update process, where all changes to data, features, or model parameters are tracked through the GitHub repository.

Major updates require re-validation of model performance and threshold calibration prior to deployment.

8.3 Model Inventory Registration

Model registration details for centralized governance.

Not Applicable to Project

---

9. Model Monitoring Strategy

9.1 Performance Monitoring

Metrics tracked:

- ROC-AUC
- Recall
- Precision
- F1 score
- Positive prediction rate

Thresholds: Performance should be reviewed if recall drops materially below historical validation levels or if precision changes significantly enough to reduce operational usefulness (i.e, more robust education

campaigns are needed as county-specific identification becomes too minimal and cases expand exponentially across the county).

Frequency: ~ Weekly, aligned with Johns Hopkins measles data updates

Recall drop >10% from validation baseline

ROC-AUC drop >5%

Feature distribution shift beyond defined statistical thresholds

9.2 Data Drift & Stability Monitoring

Drift metrics:

- Changes in feature distributions over time
- Changes in vaccination coverage distributions
- Changes in predicted positive rates
- Changes in class prevalence for both target definitions

Feature distribution changes: Special monitoring should focus on

- Not_covid_vaccinated
- Estimated hesitant
- AGE_18_44_RATIO
- AGE_CHILD_RATIO
- Social Vulnerability Index (SVI)

9.3 Alerts & Escalation Procedures

Alerts should be triggered when:

- model recall or ROC-AUC deteriorates materially
- data drift is observed in key explanatory variables
- missingness materially increases in critical source data

Escalation Procedures:

- initial review by model developer
- retraining or recalibration if degradation persists
- communication of material changes to stakeholders using dashboard outputs

9.4 Periodic Review Requirements

Quarterly / annual review expectations.

Quarterly review of:

- model performance
- threshold suitability
- feature stability
- source data quality

Annual review of:

- target definition suitability
- modeling assumptions
- feature engineering relevance
- explainability outputs

---

10. Validation Summary - NOT NECESSARY FOR PROJECT

(Completed by independent validator or internal MRM function.)

10.1 Validation Scope N/A

10.2 Validation Findings

- Strengths N/A
- Weaknesses N/A
- Remediation items N/A

10.3 Model Risk Rating N/A

10.4 Validation Conclusion / Recommendation N/A

---

11. Compliance & Regulatory Considerations - NOT NECESSARY PROJECT

- Relevant regulatory frameworks N/A

- Compliance controls N/A
- Data protection considerations (GDPR, DPDP, HIPAA, etc.) N/A

---

12. Documentation Appendix

Diagrams:

- End-to-end data pipeline
  - Ingest sources → clean and normalize → feature engineering and selection (dropping features that are correlated, contain race, or geographic data) → model training, tuning, and parameter selection → explainability and results
- Threshold selection workflow
  -
```
# ----------------------------------------
# VALIDATION: find optimal threshold
# ----------------------------------------
y_val_probs_xgb = best_xgb.predict_proba(X_val)[:, 1]
opt_threshold, best_f1 = find_best_threshold(y_val, y_val_probs_xgb)
y_val_pred_xgb = (y_val_probs_xgb >= opt_threshold).astype(int)

xgb_metrics = get_metrics(y_val, y_val_pred_xgb, y_val_probs_xgb)

print("Best XGBoost Params:", xgb_search.best_params_)
print(f"Optimal XGBoost Threshold: {opt_threshold:.4f}")
print(f"Best Validation F1 at Optimal Threshold: {best_f1:.4f}")
print("XGBoost Validation Metrics:", xgb_metrics)
```

Technical specifications:

- Final model hyperparameters
  - For Target_At_Least_1:
    - Best XGBoost Params: {'colsample_bytree': np.float64(0.733015577358303), 'learning_rate': np.float64(0.03279351625419417), 'max_delta_step': 2, 'max_depth': 3, 'n_estimators': 315, 'reg_alpha': np.float64(0.352568856334169), 'reg_lambda': np.float64(0.30478125815802903), 'subsample': np.float64(0.7493967559428825)}
    - Optimal XGBoost Threshold: 0.4599
    - Best Validation F1 at Optimal Threshold: 0.3239
    - XGBoost Validation Metrics: {'Precision': 0.22330097087378642, 'Recall': 0.5897435897435898, 'F1': 0.323943661971831, 'ROC-AUC': 0.7511103215491207}
  - For Target_At_Least_2:
    - Best XGBoost Params: {'colsample_bytree': np.float64(0.8348352022414609), 'learning_rate': np.float64(0.01954101164904113), 'max_delta_step': 3, 'max_depth': 7, 'n_estimators': 298, 'reg_alpha': np.float64(0.5912977877077271), 'reg_lambda': np.float64(0.27472179299006416), 'subsample': np.float64(0.8683730277543102)}

- Optimal XGBoost Threshold: 0.6372
- Best Validation F1 at Optimal Threshold: 0.3846
- XGBoost Validation Metrics: {'Precision': 0.37037037037037035, 'Recall': 0.4, 'F1': 0.38461538461538464, 'ROC-AUC': 0.8168232662192394}

- Feature definitions and transformations

Code snippets (non-sensitive):

- FIPS merge logic

```python
df_full = pd.merge(
    census_clean,
    agg_measles,
    on='FIPS',
    how='left'
)

df_full['total_measles_cases'] = df_full['total_measles_cases'].fillna(0)
```

- target construction logic

```python
# ========================
# 5. CREATE TARGET VARIABLES
# ========================
df_full['Target_At_Least_1'] = np.where(df_full['total_measles_cases'] >= 1, 1, 0)
df_full['Target_At_Least_2'] = np.where(df_full['total_measles_cases'] >= 2, 1, 0)
```

- feature engineering examples

```python
# ============================
# 5. FEATURE ENGINEERING
# ============================
if "Estimate (%)" in df.columns:
    df["unvaccinated_rate"] = 1 - df["Estimate (%)"]
    df["low_vax_flag"] = (df["Estimate (%)"] < 0.9).astype(int)

if (
    "Social Vulnerability Index (SVI)" in df.columns
    and "Estimated hesitant" in df.columns
):
    df["risk_interaction_2"] = (
        df["Social Vulnerability Index (SVI)"] * df["Estimated hesitant"]
    )

if (
    "Percent adults fully vaccinated against COVID-19 (as of 6/10/21)" in df.columns
    and "Estimated hesitant" in df.columns
):
    df["vax_hesitancy_interaction"] = (
        df["Percent adults fully vaccinated against COVID-19 (as of 6/10/21)"]
        * df["Estimated hesitant"]
```

- threshold optimization logic

```python

def find_best_threshold(y_true, y_probs):
    precisions, recalls, thresholds = precision_recall_curve(y_true, y_probs)

    if len(thresholds) == 0:
        return 0.5, 0.0
    f1_scores = (2 * precisions[:-1] * recalls[:-1]) / (
        precisions[:-1] + recalls[:-1] + 1e-10
    )
    best_idx = np.argmax(f1_scores)
    return thresholds[best_idx], f1_scores[best_idx]
```

Glossary:

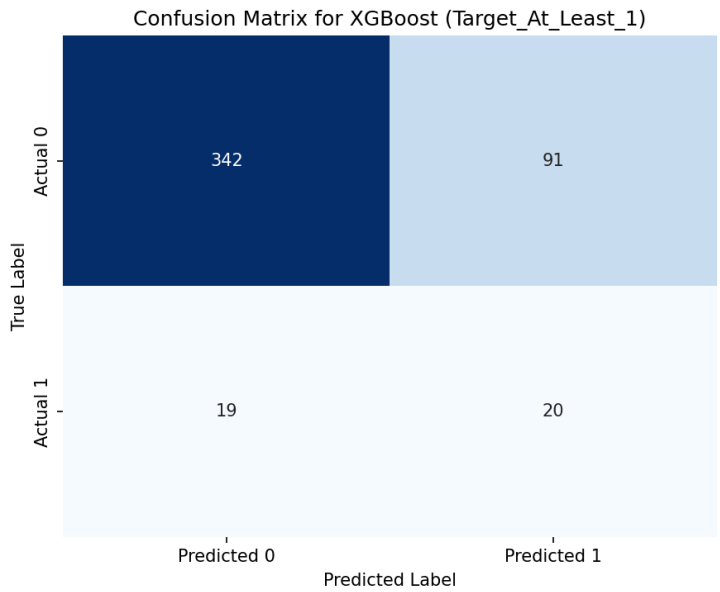
- ROC-AUC: ability for a model to distinguish between two classes (1 would be perfect, .5 is random, less than 0.5 the model is flawed)
- Precision: measures the ability to accurately classify a case that is positive as positive
- Recall: Proportion of actual positive cases correctly identified by the model
- SHAP: provides local explainability for XGBoost model
- SVI: measures a community's susceptibility to negative impacts from disasters or crises ranging from 0 (lowest) to 1 (highest)

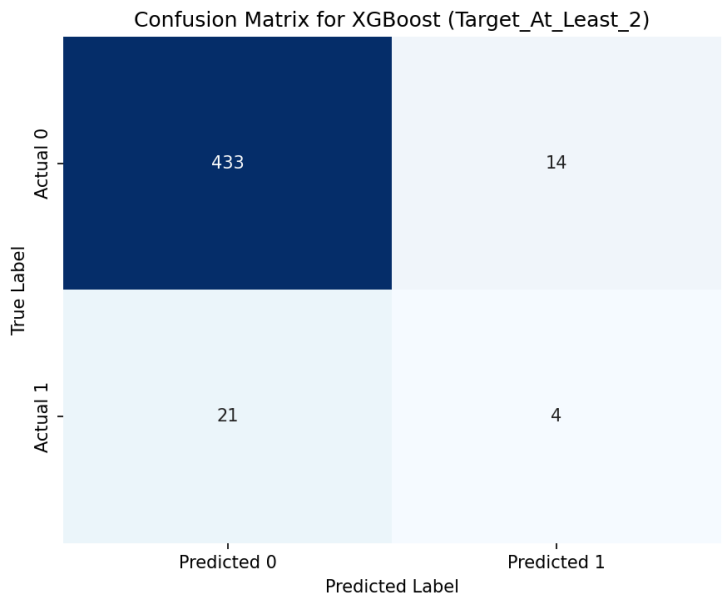
Acronyms:

- CDC: Centers for Disease Control and Prevention
- SVI: Social Vulnerability Index
- MMR: Measles Mumps and Rubella Vaccine
- SHAP:SHapley Additive exPlanations

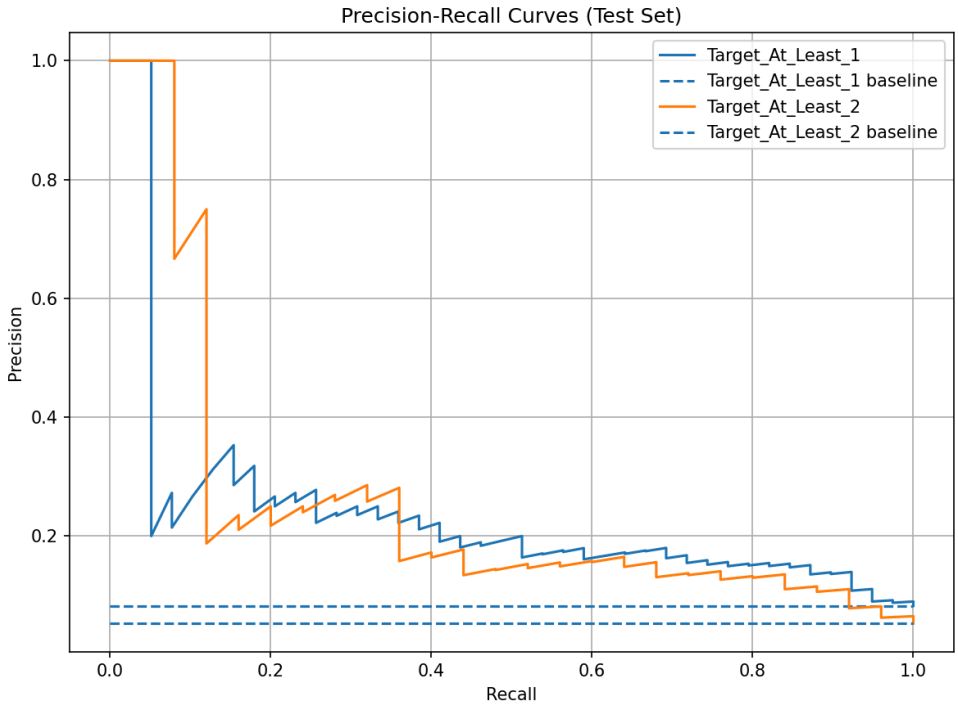
Testing logs:

- validation performance summaries
- test set confusion matrices





- precision-recall curves



Decision logs:

- removal of clustering from final project scope
- comparison of Target_At_Least_1 versus Target_At_Least_2
- selection of XGBoost as primary model